

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1837

Pensacola Florida Plant 138 kV
March 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1837. The Interconnection Customer proposes a 75.5 MW solar photovoltaic facility interconnecting at the Pensacola Florida Plant 138 kV substation in the Maryland area. The request was received on July 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 515 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$87,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FULLERTON / WINDY EDGE STTION 138.0 kV	N STAUNTON / TANGUISSON 138.0 kV	107.6	269	6.9
TANGUISSON / N STAUNTON 138.0 kV line	TANGUISSON / FULLERTON 138.0 kV	103.7	437	5.9
TANGUISSON / RAPHAEL RD 138.0 kV Cir. 1	RAPHAEL ROAD / TANGUISSON 138.0 kV #2	101.8	509	6.0
RAPHAEL RD / TANGUISSON 138.0 kV L	TANGUISSON / RAPHAEL RD 138.0 kV Ckt 1	101.8	509	6.0
JOPPATOWNE / RAPHAEL ROAD 138.0 kV Line 1	JOPPATOWNE / RAPHAEL ROAD 138.0 kV Circuit 2	101.6	513	5.8
JOPPATOWNE SUBSTATION / RAPHAEL ROAD 138.0 kV Cir. 2	JOPPATOWNE / RAPHAEL RD 138.0 kV Line	101.6	513	5.8
FULLERTON / TANGUISSON 138.0 kV	N STAUNTON / TANGUISSON 138.0 kV line	100.0	488	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FULLERTON / WINDY EDGE STTION 138.0 kV	Reconductor line	\$752,000	2% (\$18,000)
NORTH STAUNTON / TANGUISSON 138.0 kV L	Reconductor line	\$387,000	2% (\$6,000)
TANGUISSON SUBSTATION / RAPHAEL RD 138.0 kV	Reconductor line	\$1,306,000	1% (\$18,000)
TANGUISSON / RAPHAEL ROAD 138.0 kV #2	Reconductor line	\$1,040,000	1% (\$15,000)
RAPHAEL RD / JOPATOWNE SUBSTATION 138.0 kV	Reconductor line	\$928,000	1% (\$13,000)
JOPPATOWNE SUBSTATION / RAPHAEL ROAD 138.0 kV Cir. 2	Reconductor line	\$873,000	1% (\$12,000)
FULLERTON / TANGUISSON 138.0 kV	Reconductor line	\$375,000	1% (\$5,000)
	Total allocated to project		\$87,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.